

Mechanistic insights in the arylation of $[\text{FeCp}(\text{CO})_2\text{X}]$ complexes using Zn/Cu/Au derivatives. Influence of the metallic counterpart.

Miguel Claros,^a Guillermo Marcos-Ayuso^a and Juan A. Casares^a

^aIU CINQUIMA/Química Inorgánica, Facultad de Ciencias, Universidad de Valladolid, 47011-Valladolid, Spain

e-mail: miguel.claros@uva.es; juanangel.casares@uva.es

Keywords: Transmetallation, Mechanistic insights, Iron complexes, Photochemistry, Arylation.

Eighteen-electrons organometallic complexes of the type $[\text{M}^{\text{II}}\text{Cp}]$ can follow mainly two different activation pathways to be enrolled in a catalytic process. Firstly, and more common, is a ligand dissociation to give a more reactive sixteen-electrons intermediate. On the other hand, a $^5\eta$ to $^3\eta$ process can generate an electron vacancy which allow association mechanisms. In the case of organometallic iron complexes, it is well accepted that follows mainly dissociative mechanisms throw sixteen-electrons species. This kind of complexes have been employed during the las decades in important catalytic processes like hydrosilylation of aldehydes or ketones and hydrophosphination reactions.^[1]

Whether such processes take place after a first dissociative step, it is crucial to understand this dissociation to improve these catalytic systems. Different studies with $[\text{FeCp}(\text{CO})_2\text{X}]$ complexes (where X = Cl, Br, I) reveals that a CO molecule can dissociate by four different pathways among which are chemically, photochemically, thermally, and metal catalyzed throw a 16 e⁻ transition states. This dissociative step can be followed by different transmetallation reactions from organogold,^[2] organozincates^[3] or copper complexes (Figure 1).^[4]

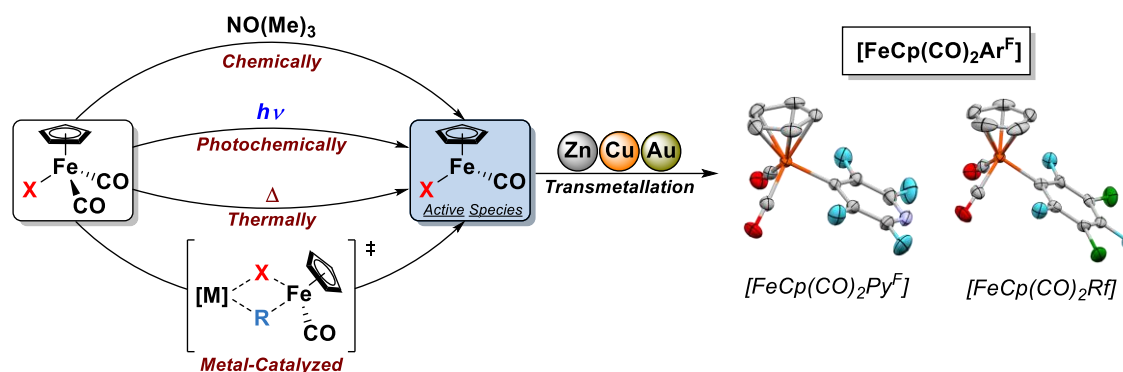


Figure 1. Plausible mechanisms for the dissociative step that precedes the transmetallation of $[\text{FeCp}(\text{CO})_2\text{X}]$ complexes.

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